

# Graham Positivity of Quantum Double Schubert Polynomials

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## Abstract

We prove a positivity property for the coefficients in the product expansion of quantum double Schubert polynomials. The proof relies on the refined Graham positivity theorem.

## 1 Introduction

Let  $S_\infty = \bigcup_{m \geq 1} S_m$ ,  $x = (x_1, x_2, \dots)$ ,  $y = (y_1, y_2, \dots)$ ,  $t = (t_1, t_2, \dots)$ , and  $q = (q_1, q_2, \dots)$ . Consider the following product expansion:

$$\mathfrak{S}_u^q(x; y) \mathfrak{S}_v^q(x; t) = \sum_{w \in S_\infty} c_{u,v}^{w,q}(y, t; q) \mathfrak{S}_w^q(x; t). \quad (1)$$

We have the following conjecture:

**Conjecture 1.1.** *For any  $u, v, w \in S_\infty$ ,  $c_{u,v}^{w,q}(y, t; q) \in \mathbb{N}[t_i - y_j, q_k]_{i,j,k \geq 1}$ .*

In particular, when  $q = 0$ , the conjecture reduces to the result proved by Yibo Gao et al. [3, Theorem 1.1].

We first briefly describe the proof of the conjecture. Fix  $n \geq 1$ , and let  $G = GL_{2n}(\mathbb{C})$  and  $X = G/B = \text{Fl}_{2n}$ . Let  $B^-$  be the opposite Borel subgroup and  $T = B \cap B^-$ . For  $w \in S_{2n}$ , write

$$X_w = \overline{B^- w B / B}, \quad Y_w = \overline{B w B / B}.$$

Embed  $S_n$  into  $S_{2n}$  so that it fixes the last  $n$  letters. Let

$$\tau(i) = n + i, \quad \tau(n + i) = i, \quad a = (a_1, \dots, a_{2n}) = (t_1, \dots, t_n, y_1, \dots, y_n).$$

For an effective degree  $d \in \mathbb{Z}_{\geq 0}^{2n-1}$ , let  $M_d = \overline{M}_{0,3}(X, d)$ , and let  $\text{ev}_i : M_d \rightarrow X$  be the evaluation map. Define

$$c_{u,v}^{w,d;\tau}(a) = (\pi_{M_d})_*^T (\text{ev}_1^*[\tau X_u]^T \text{ev}_2^*[X_v]^T \text{ev}_3^*[Y_w]^T \cap [M_d]^T). \quad (2)$$

**Fact 1.1.** *When  $n$  is sufficiently large, the coefficients in (1) satisfy*

$$c_{u,v}^{w,q}(y, t; q) = \sum_d c_{u,v}^{w,d;\tau}(a) q^d,$$

**Claim 1.1.** *For every effective degree  $d$ ,*

$$c_{u,v}^{w,d;\tau}(a) \in \mathbb{N}[a_i - a_{n+j}]_{1 \leq i,j \leq n} = \mathbb{N}[t_i - y_j]_{1 \leq i,j \leq n}.$$

*Proof of Conjecture 1.1.* Take  $n$  sufficiently large. By Fact 1.1 and Claim 1, we have:

$$c_{u,v}^{w,q} = \sum_d c_{u,v}^{w,d;\tau} q^d \in \mathbb{N}[a_i - a_{n+j}, q_k] = \mathbb{N}[t_i - y_j, q_k]$$

□

## 2 Proof of Claim

Let  $N^-$  be the unipotent radical of  $B^-$ , let  $w_0^{(r)}$  be the longest element of  $S_r$ , and write

$$N^-(w) = N^- \cap wN^-w^{-1}, \quad B^-(w) = TN^-(w).$$

For a fixed effective degree  $d$ , define

$$I_{u,v}^d = (\text{ev}_1, \text{ev}_2)^{-1}(\tau X_u \times X_v), \quad \Gamma_{u,v}^d = (\text{ev}_3)_*[I_{u,v}^d].$$

We will use the following facts.

**Fact 2.1.** *Let  $a_0 = 1 \times w_0^{(n)} \in S_{2n}$ . If  $u, v \in S_n$ , then*

$$a_0 X_u = X_u, \quad a_0 X_v = X_v, \quad a_0^{-1} \tau a_0 = w_0^{(2n)}.$$

**Fact 2.2** ([7], Lemma 6.3; [5, 8]; [7], formula (2')). *Let  $F : Z \rightarrow X \times X$  be equivariant for the diagonal  $G$ -action, and suppose that  $Z$  is irreducible. Then  $F^{-1}(Y_p \times X_q)$  is reduced and locally irreducible, with codimension  $\dim X - \ell(p) + \ell(q)$ .*

(i) *When  $X = G/P$ , the stable map space  $\overline{M}_{0,r}(X, d)$  is irreducible.*

(ii) *Let  $f : X_1 \rightarrow X_3$  be a morphism between projective  $T$ -varieties, where  $X_3$  is smooth and  $V \subset X_3$  is a  $T$ -invariant subvariety of codimension  $c$ . If the irreducible components  $V_i$  of  $f^{-1}(V)$  all have codimension  $c$  in  $X_1$  and occur with algebraic multiplicity  $m_i$ , then*

$$f_T^*[V]_T \cap [X_1]_T = \sum_i m_i [V_i]_T.$$

**Fact 2.3** ([7], Proposition 3.1). *For the structure map  $\pi_X : X \rightarrow \text{pt}$ , one has:*

$$(\pi_X)_*^T([X_z]^T[Y_w]^T) = \delta_{z,w}, \tag{3}$$

**Fact 2.4** ([7], Section 3.1, formulas (1) and (2)). *Let  $f : Z_1 \rightarrow Z_2$  be a projective  $T$ -equivariant morphism. Its equivariant Gysin pushforward  $f_*^T : H_*^T(Z_1) \rightarrow H_*^T(Z_2)$  satisfies:*

(i) *If  $g : Z_2 \rightarrow Z_3$  is also a projective  $T$ -equivariant morphism, then  $(g \circ f)_*^T = g_*^T \circ f_*^T$ ;*

(ii) *For  $\alpha \in H_*^T(Z_2)$  and  $\beta \in H_*^T(Z_1)$ , one has  $f_*^T(f^* \alpha \cap \beta) = \alpha \cap f_*^T(\beta)$ ;*

(iii) *For the fundamental class of an algebraic cycle  $V$ , one has  $f_*^T[V]^T = [f_* V]^T$ ,*

**Fact 2.5** ([3], Theorem 2.3 and Corollary 2.4). *Let  $G$  be a connected complex reductive group and  $B^-$  an opposite Borel subgroup. For a Weyl group element  $w$ , let  $B^-(w) = T(N^- \cap wN^-w^{-1})$ , and let  $I(w) = \{\alpha \in \Phi^+ : w^{-1}\alpha \in \Phi^-\}$ . If  $Y$  is an effective  $B^-(w)$ -invariant cycle on a smooth  $B^-$ -variety, then its equivariant class can be written as*

$$[Y]^T = \sum_r f_r [Z_r]^T, \quad f_r \in \mathbb{N}[-\alpha]_{\alpha \in I(w)},$$

where each  $Z_r$  is an effective  $B^-$ -invariant cycle. In particular, when the target is  $G/B$ , each  $Z_r$  is a nonnegative integer linear combination of opposite Schubert cycles, and hence

$$[Y]^T = \sum_{z \in W} p_z [X_z]^T, \quad p_z \in \mathbb{N}[-\alpha]_{\alpha \in I(w)}.$$

Using these facts, we prove Claim 1.1.

*Proof of Claim 1.1.* Define the following two maps:

$$F = (\text{ev}_1, \text{ev}_2) : M_d \longrightarrow X \times X$$

$$L_g : X \longrightarrow X, \quad L_g(xB) = gxB.$$

Let  $g = a_0^{-1} \in G$ , and consider the left action:

$$\Psi_g : M_d \rightarrow M_d, \quad \Psi_g(C, p_1, p_2, p_3, f) = (C, p_1, p_2, p_3, g \circ f).$$

Thus, for any  $m = (C, p_1, p_2, p_3, f) \in M_d$ , we have

$$\begin{aligned} (F \circ \Psi_g)(m) &= (gf(p_1), gf(p_2)) \\ &= (L_g(\text{ev}_1(m)), L_g(\text{ev}_2(m))) \\ &= ((L_g \times L_g) \circ F)(m). \end{aligned}$$

Hence  $F \circ \Psi_g = (L_g \times L_g) \circ F$ .

By Fact 2.1,  $I_{u,v}^d \cong F^{-1}(Y_{w_0u} \times X_v)$ , and by Fact 2.1,  $g(\tau X_u) = Y_{w_0u}$  and  $g(X_v) = X_v$ , therefore,

$$(L_g \times L_g)(\tau X_u \times X_v) = Y_{w_0u} \times X_v.$$

Since  $M_d$  is irreducible, Fact 2.2 gives:

$$\text{codim}_{M_d} I_{u,v}^d = \dim X - \ell(w_0u) + \ell(v) = \ell(u) + \ell(v).$$

Since  $\ell(w_0u) = \dim X - \ell(u)$ , it follows that  $I_{u,v}^d$  is reduced and locally irreducible. By the pullback formula in Fact 2.2,

$$[I_{u,v}^d]^T = \text{ev}_1^*[\tau X_u]^T \text{ev}_2^*[X_v]^T \cap [M_d]^T.$$

Since  $\pi_{M_d} = \pi_X \circ \text{ev}_3$ , we have

$$c_{u,v}^{w,d;\tau}(a) = (\pi_X)_*([Y_w]^T \cap (\text{ev}_3)_*[I_{u,v}^d]^T) = (\pi_X)_*([I_{u,v}^d]^T[Y_w]^T).$$

We next show that  $\Gamma_{u,v}^d$  is an effective  $B^-(\tau)$ -invariant cycle and that

$$[\Gamma_{u,v}^d]^T = \sum_{z \in S_{2n}} p_{u,v,d}^z(a) [X_z]^T, \quad p_{u,v,d}^z(a) \in \mathbb{N}[t_i - y_j]_{1 \leq i, j \leq n}. \quad (4)$$

Indeed,  $X_v$  is  $B^-$ -invariant, and if  $h = \tau h' \tau^{-1} \in N^- \cap \tau N^- \tau^{-1}$ , then

$$h(\tau X_u) = \tau h' X_u \subseteq \tau X_u$$

Thus  $I_{u,v}^d$  is  $B^-(\tau)$ -invariant.

Since  $I_{u,v}^d$  is a reduced closed subscheme of  $M_d$ ,  $[I_{u,v}^d]$  is an effective cycle, and every irreducible component has algebraic multiplicity 1. On the other hand,  $X$  is projective, so  $M_d$  is proper. Therefore  $\text{ev}_3 : M_d \rightarrow X$  is proper, and hence  $\Gamma_{u,v}^d$  is an effective cycle on  $X$ .

Since  $\text{ev}_3$  is  $G$ -equivariant and  $I_{u,v}^d$  is  $B^-(\tau)$ -invariant,  $\Gamma_{u,v}^d$  is also  $B^-(\tau)$ -invariant. Therefore, Fact 2.5 gives:

$$[\Gamma_{u,v}^d]^T = \sum_{z \in S_{2n}} p_{u,v,d}^z(a) [X_z]^T, \quad p_{u,v,d}^z(a) \in \mathbb{N}[t_i - y_j]_{1 \leq i, j \leq n}.$$

The inversion set is

$$I(\tau) = \{\epsilon_i - \epsilon_{n+j} : 1 \leq i, j \leq n\} = \{y_j - t_i : 1 \leq i, j \leq n\}$$

Finally, by Fact 2.3,

$$c_{u,v}^{w,d;\tau}(a) = (\pi_X)_*^T \left( \sum_{z \in S_{2n}} p_{u,v,d}^z(a) [X_z]^T [Y_w]^T \right) \quad (5)$$

$$= \sum_{z \in S_{2n}} p_{u,v,d}^z(a) \delta_{z,w} = p_{u,v,d}^w(a). \quad (6)$$

Therefore,

$$c_{u,v}^{w,d;\tau}(a) = p_{u,v,d}^w(a) \in \mathbb{N}[t_i - y_j]_{1 \leq i, j \leq n}. \quad (7)$$

This proves Claim 1.1.  $\square$

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